

# TrayGuard: What We Learned About Why Tray Errors Persist

EC ENGR 180DW — Final Design Review

Owen Fan, Matthew Li, Andy Ma, Jacob Moore, Qiyu Xin

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# Roadmap

- 1 The Problem
- 2 Why Tray Errors Persist
- 3 CV Attempt
- 4 Why CV Alone Won't Work
- 5 Why Training Is the Best Entry Point
- 6 What a Future Team Should Do
- 7 Conclusion

# The Problem

# The Problem By The Numbers

**9 per 100 cases** — Alfred et al.  
[1] 3,900 defects in 41,799 cases  
**55% at assembly**

**10 min avg delay** — Nichol et al. [2] 236 errors, 147 cases  
**\$6–9M/yr estimated cost**

**44% wrong spec** — Zhu et al. [3] 398 errors in 33,839 packages  
**largest single category**

These are recurring, measured operational failures — not isolated accidents.

# Why Tray Errors Persist

# Lookalike Instruments: Kelly vs Crile

## Kelly and Crile forceps comparison



Kelly and Crile forceps: Both have transverse serrations. Used for medium to large vessels. Crile's serrations along entire length of jaws. Kelly's serrations only on ½ of the distal jaws

## Kelly Forceps sizing



Kelly and Crile: same family, different serration patterns. Sizes 6", 7", 8" — a staff might mix 14cm and 16cm clamps. Total count correct, specification wrong. [4]

# What Tray Reconstruction Actually Requires



SPD assembly workstation [5]

- Identify each instrument by sight (no labels)
- Match against a count sheet with **local names**
- Distinguish lookalikes (straight vs curved Mayo)
- Check for damage, wear, contamination
- Count correct quantities
- All under **production pressure**

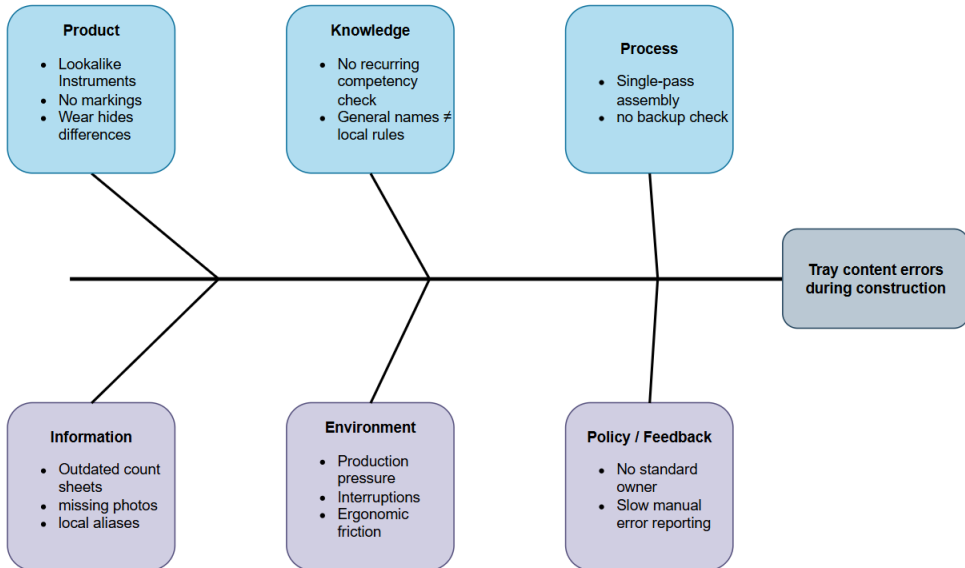
# Common Surgical Instruments

| Instrument                                                                        | Manufacturer / Formal Name                                         | Common SPD / OR Name (Local Name)                                                                            |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  | <b>Rochester-Ochsner Forceps</b><br>(Curved, 1x2 teeth)            | <b>Kocher Clamp</b>       |
|  | <b>Halsted Mosquito Forceps</b><br>(Straight, 1x2 teeth)           | <b>Mosquito Clamp</b>     |
|  | <b>Foerster Sponge Forceps</b><br>(Serrated tips)                  | <b>Sponge Stick</b>       |
|  | <b>Schnidt Tonsil Forceps</b><br>(Curved, serrated)                | <b>Tonsil Clamp</b>       |
|  | <b>Mayo-Hegar Needle Holder</b><br>(With tungsten carbide inserts) | <b>Needle Driver</b>      |
|  | <b>Backhaus Towel Clamp</b><br>(For drapes and towels)             | <b>Towel Clip</b>         |
|  | <b>Adson Tissue Forceps</b><br>(1x2 teeth)                         | <b>Rat-Tooth Forceps</b>  |
|  | <b>Dressing Forceps</b><br>(Smooth, straight)                      | <b>Smooth Forceps</b>     |
|  | <b>Metzenbaum Scissors</b><br>(Curved)                             | <b>Metz</b>               |
|  | <b>Richardson Retractor</b><br>(Double-ended)                      | <b>Rich Retractor</b>     |

Source: ChatGPT 5.5



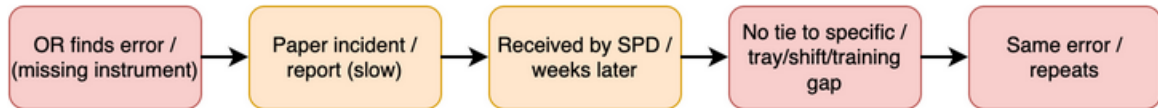
# Root Cause Analysis



# Root Cause Analysis

- ① SPD evolved from materials management → **no licensure, weak professionalization**
- ② Cost-center accounting → wage investment is visible, error cost is invisible
- ③ No information architecture → downstream OR errors don't trigger upstream correction

# The Broken Feedback Loop



*"Error reporting is cumbersome, human-dependent, delayed, and incomplete." — Nichol et al.*

CV Attempt

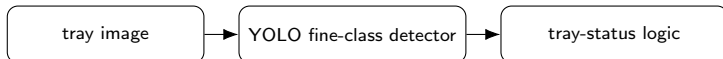
# First CV Attempt: One Detector Did Everything

## 1-stage pipeline

- 1 Overhead image of tray
- 2 YOLO detects every fine class directly
- 3 Count detections against the tray list
- 4 Flag missing, wrong, extra, or uncertain items

## Why this was brittle

- Localization and fine ID were coupled
- Similar tools shared shape and metal texture
- One wrong subtype became a tray error
- Overlay and lighting shifts reduced reliability



# The Problem Was Fine-Grained Confusion, Not Just Detection

## Flat fine labels

| Fine class                | Visual family |
|---------------------------|---------------|
| Scalpel n4                | scalpel       |
| Straight dissection clamp | clamp         |
| Straight Mayo scissor     | scissors      |
| Curved Mayo scissor       | scissors      |

## What the tests showed

- Clean matched data: high mAP was achievable
- Layout shift: separated → overlay dropped sharply
- Real instruments: per-class AP depended on split
- Lookalikes needed a narrower decision context

The detector should first answer **“what family is this?”** The classifier should answer **“which subtype?”**

# Second CV Attempt: Coarse-to-Fine 2-Stage Pipeline

## Stage 1: general detector

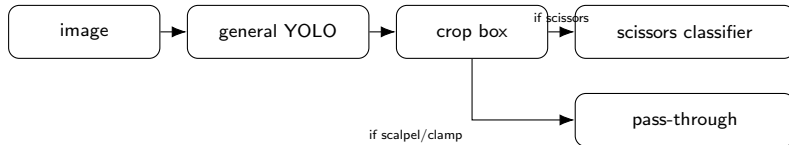
YOLO predicts broad families only:

- scalpel
- clamp
- scissors

## Stage 2: specific classifier

Only runs when a family has variants:

- scissors → straight vs curved
- scalpel → pass-through
- clamp → pass-through



# Testing the 2-Stage Demo on Separated Images

## Dataset builder

- Remap fine labels into general classes
- Reuse ground-truth boxes for YOLO labels
- Crop boxes into ImageFolder classifier data
- Train one selected specific classifier

## End-of-training

- Run on all separado\*.jpg images
- Draw general boxes and confidence
- Classify cropped scissors as straight/curved
- Export annotated images, crops, CSV, HTML report

## What the training shows

General detection and fine-grained scissors classification can be tested together while keeping their failure modes separate.



## 2-Stage Helped, But Scissor ID Still Failed

### Helped

*[figure placeholder]*

- Broad family detection worked.
- Scissors crops routed to classifier.

### Still failed

*[figure placeholder]*

- Both scissors predicted as `Straight_Mayo`.
- Fine-grained lookalike ID remained unreliable.

### Takeaway

The 2-stage design separated detection from subtype classification, but it was still not reliable enough for deployment.

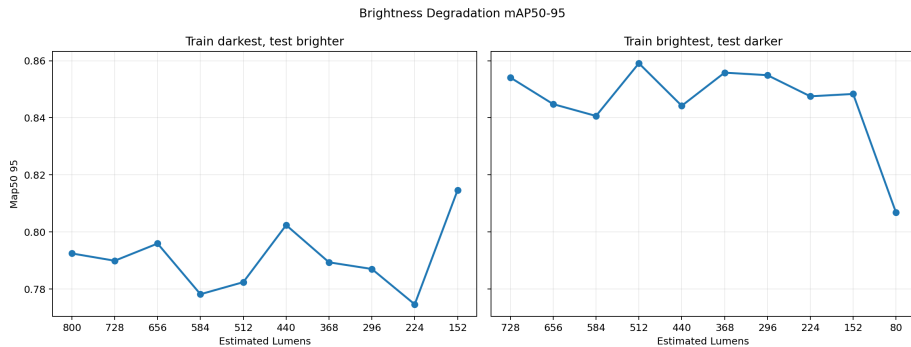
# Why CV Alone Won't Work

# We Ran 7 Staged Detection Experiments

| Experiment        | What it tested      | Key result                |
|-------------------|---------------------|---------------------------|
| Bright→dim        | Light tolerance     | mAP50-95 = 0.781          |
| Matte→reflective  | Background transfer | mAP50-95 = 0.759          |
| Separated→overlay | Occlusion           | <b>mAP50-95 = 0.389</b>   |
| Real instruments  | Shape vs position   | Per-class AP: 0.000–0.862 |

Lavado baseline (clean data): **mAP50 = 0.990**. Pipeline works — the data doesn't generalize.

# Brightness Degradation



# Occlusion Is the Dealbreaker

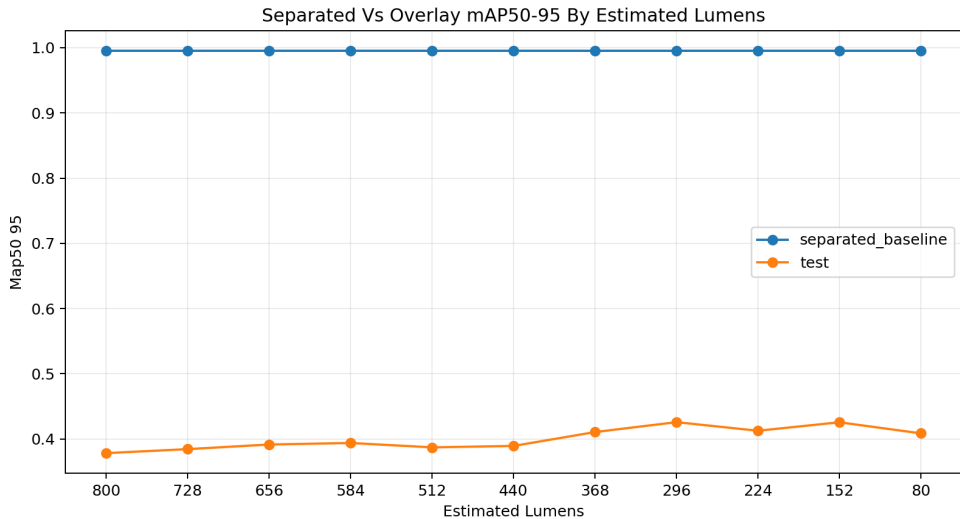
**Separated layout (training)**



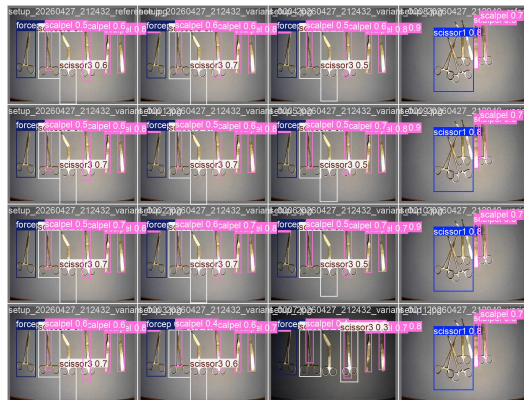
**Overlay layout (test)**



# Overlay Degradation



# Real Instruments: The Model Learned Position, Not Shape



| Split            | Surviving classes         | Zero AP classes        |
|------------------|---------------------------|------------------------|
| Matte→reflective | scissor2, forcep, scalpel | scissor1/3/4           |
| Reflective→matte | <b>scalpel only</b>       | scissor1/2/3/4, forcep |
| Shape similarity | scissor1, scissor4        | scissor2/3, forcep     |

# The Cumulative Lesson: CV Validation Burden

## What a deployment-grade CV system still needs

- ✗ Site-specific training data (instruments, lighting, layouts)
- ✗ Cross-manufacturer validation [6] (Kienle: major performance drop)
- ✗ Open-set rejection (unknown objects = false positives)
- ✗ Human-in-the-loop for every uncertain case
- ✗ Workflow integration in live SPD (never tested)
- ✗ Regulatory / device policy / privacy review

The training path doesn't need any of this for a first study.



# We Evaluated Seven Solution Categories

| Solution                   | Buildable?         | Validatable? | Creates infra for future?  |
|----------------------------|--------------------|--------------|----------------------------|
| CV tray checker            | No (needs SPD)     | No           | Maybe                      |
| Physical error-proofing    | No (custom fab)    | No           | No                         |
| Workflow redesign          | No (policy change) | No           | No                         |
| Decision support           | Partial            | Partial      | Yes (local content)        |
| Task simplification        | No (hospital data) | No           | No                         |
| Accountability system      | No (org policy)    | No           | No                         |
| <b>Training (selected)</b> | <b>Yes</b>         | <b>Yes</b>   | <b>Yes (module schema)</b> |

# Why Training, Not Because Training Is a Cure

## What training **addresses**

- Measurable local tray knowledge
- Lookalike discrimination practice
- Pre/post learning evidence

## What training does **NOT** fix

- Broken count sheets
- Staffing shortages
- Structural wage problems
- The broken feedback loop

Training is the one piece we can build and validate **now**.

# Why Training Is the Best Entry Point

# CV and Training Are Complementary

FIGURE TBD

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The module is the durable artifact. The training loop proves the module works. The module then enables CV.

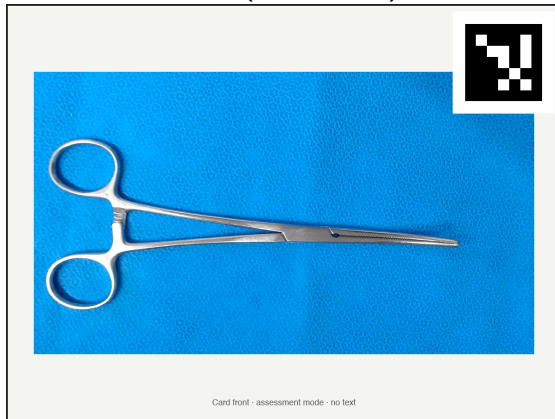
# The Durable Design Artifact: Tray Module Schema

```
{
  "id": "mayo_scissors_straight_55",
  "display_name": "Straight Mayo, 5.5 in",
  "aliases": ["straight Mayo",
              "suture scissors"],
  "distinguishing_features": [
    "Heavy straight blades"
  ],
  "common_confusions": [
    "mayo_scissors_curved_55"
  ],
  "apriltag_id": 12,
  "study_prompts": [ ... ]
}
```

- Mirrors real count-sheet fields
- Adds aliases, lookalikes, prompts
- File-backed, versioned
- **Software is throwaway.**
- **Schema survives.**

# Printed Cards: How It Works Physically

## Front (assessment)



## Back (study)

### Clamp Kelly 8"

Family: Clamping/occluding · Catalog: CLAMP KELLY 8"

#### Key identification features

- Curved jaws with ring handles
- Transverse serrations on distal half of jaws
- 8-inch overall length

#### Common lookalikes

- Crile clamp — serrations run full jaw (Kelly: distal half only)
- Kocher clamp — single tooth at tip (Kelly: no tooth)
- Kelly 6" / Kelly 12" — same jaw shape, different length

Also known as: JARIT\_106-191

Source: FGVC12 shared Google Drive major tray image, class JARIT\_106-191: CLAMP KELLY 8"

Neutral front prevents answer leakage. Tag ID maps to instrument ID.

# The Scoring Architecture: 5 Error Categories

| Category      | Definition                 | Example                    |
|---------------|----------------------------|----------------------------|
| Correct       | Right item, right count    | 1 scalpel handle ✓         |
| Missing       | Required item not selected | No Kelly clamp             |
| Wrong         | Item not on tray list      | Retractor not in this tray |
| Extra         | Distractor selected        | Item from different tray   |
| Misidentified | Right class, wrong variant | Curved instead of straight |
| Wrong count   | Right item, wrong quantity | 2 Kellys, need 4           |

- Maps to literature error categories [1, 3, 2]
- Can separate *what* improved (fewer misses? fewer lookalikes?)

## FIGURE TBD

figures/confidence`mockup`placeholder.pdf

### High-confidence error

Learner was sure but wrong. Targeted feedback needed. — Kruger & Dunning: "unskilled and unaware of it" [7]

### Low-confidence correct

Learner was unsure but right. Fragile knowledge, needs reinforcement. — Dunlosky et al.: metacognitive calibration predicts retention [8]



# The Learning Loop (7 Steps)

FIGURE TBD

[figures/learning`loop`placeholder.pdf](#)

Pre-test (baseline) → Study cards → Quiz → Practice sort → Post-test → Export

# The Research Question

How can a local tray training and assessment system help novice users improve their ability to detect missing, wrong, extra, misidentified, or miscounted instruments before supervised SPD tray-reconstruction work?

That question is narrow, testable, and honest about what we can claim.

What a Future Team Should Do

# The Study Ladder

FIGURE TBD

figures/study'ladder'placeholder.pdf

Kirkpatrick Level 2 target — Arthur et al.: learning gain is the standard training evaluation target [9].  
Level 3–4 require future validation.

# Study 1: Novice Pre/Post Design

| Segment            | Time      | Activity                   |
|--------------------|-----------|----------------------------|
| Consent + intake   | 3–5 min   | Background                 |
| Pre-test tray sort | 5–8 min   | No hints, no feedback      |
| Retrieval cards    | 10–12 min | Prompt-before-reveal       |
| Quiz               | 5–8 min   | With confidence            |
| Practice sort      | 8–10 min  | Error feedback             |
| Post-test          | 5–8 min   | Parallel variant, no hints |
| Photo-ID test      | 5–7 min   | Convergent validity        |

**n = 8–12** novices, one 60-min session, within-subject

## Study 2: Delayed Retention (The Falsification Criterion)

FIGURE TBD

figures/retention'placeholder.pdf

**Criterion 1** —  $\geq 50\%$  of gain retained

**Criterion 2** — Accuracy  $\geq 5pp$  above baseline

**Criterion 3** —  $\geq 70\%$  return rate

If these fail, the claim reverts to "immediate learning only."

## Study 3: Expert Review (Closing the Content Gap)

- Q Are instrument names and aliases correct for local practice?
- Q Are counts and distractors plausible for training?
- Q Are lookalike pairs educationally meaningful?
- Q Are feedback messages misleading or safe?
- Q Would the weak-item report help target coaching?

The main weakness of a student-built module is **content validity**. Expert review closes that gap.

# Go/No-Go Decision for Future Team

FIGURE TBD

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Study passes → SPD pilot is justified. Study fails → problem may be harder than training alone.



# Conclusion

# What We Know and What We Don't

## What we know

- Tray errors are systemic, recurring, and well-documented
- CV alone can't solve this — the validation burden is too high
- Training is the buildable entry point
- The module schema is durable infrastructure
- The study ladder has falsification criteria

## What we can't claim (yet)

- Training transfers to real SPD work
- System works for experienced technicians
- It reduces OR delays
- CV is deployment-ready
- Same-day gains persist (Study 2 tests this)

# The Honest Handoff

Our project found that this problem is harder and more systemic than any single technical fix can address. Our contribution is documenting *why*, and giving the next team a validated starting point — not a finished product.

The study is designed. The schema is defined. The scoring is tested. **Run the study.**

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